Winning the Space Race with Data Science

SpaceX Falcon 9 first-stage landing prediction | Capstone technical report

From raw launch records to interactive analytics and predictive modelling

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GitHub project: github.com/eokorouka-jpg/SpaceX-Falcon9-Data-Science-Capstone

Executive summary

What we investigated and what we learned

- Goal: estimate the likelihood that a Falcon 9 first stage lands successfully.
- Prepared 90 labelled launches; 60 successful and 30 unsuccessful (66.7% success).
- Explored flight history, orbit, payload, launch site, geography and booster outcomes.
- Compared four tuned classifiers on 18 held-out flights.
- Best observed test accuracy: 83.33% — Logistic Regression, SVM and KNN (tie).

Business problem and research questions

- Reusable boosters can reduce launch costs, so predicting landing outcomes matters.
- Which launch sites, orbits and payload ranges are associated with landing success?
- How do outcomes change over flight history?
- Can supervised learning predict a landing from pre-landing launch characteristics?
- How reliably do results generalize beyond this small historical sample?

End-to-end data science journey

Evidence across collection, wrangling, exploration and modelling

API / Wikipedia

Clean + label

SQL + EDA

Folium + Dash

ML + evaluation

Python • pandas • SQL • seaborn/matplotlib • Folium • Plotly Dash • scikit-learn

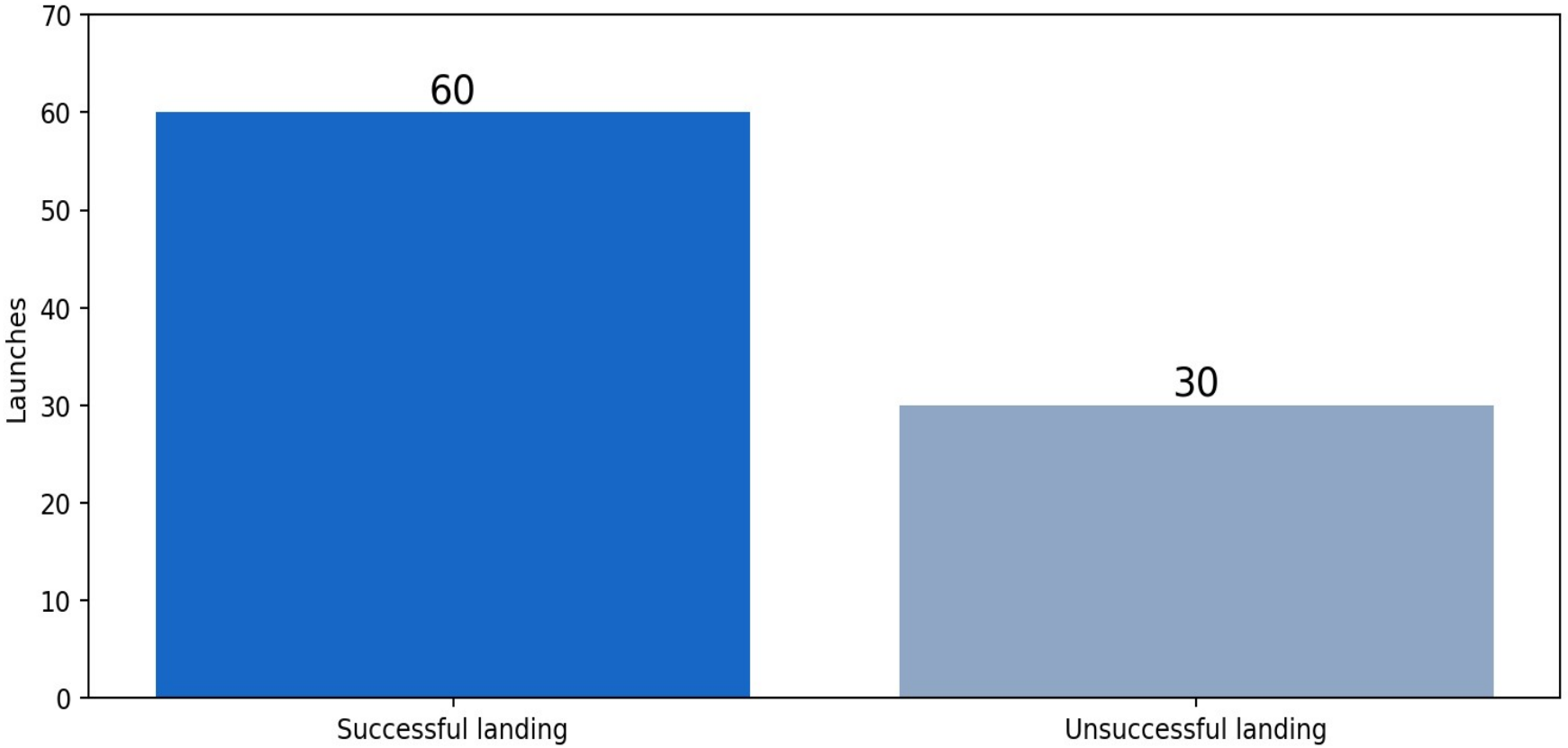
Data collection: SpaceX API and Wikipedia

- API lab: retrieve SpaceX launch JSON and expand rocket, payload, core and launchpad references.
- Wikipedia lab: use BeautifulSoup to extract Falcon 9 launch-table information.
- The saved web-scraping notebook shows an HTTP 403 response in one run; do not treat that blocked attempt as successful scraping.
- Data sources and intermediate CSV files should be committed with their notebooks.

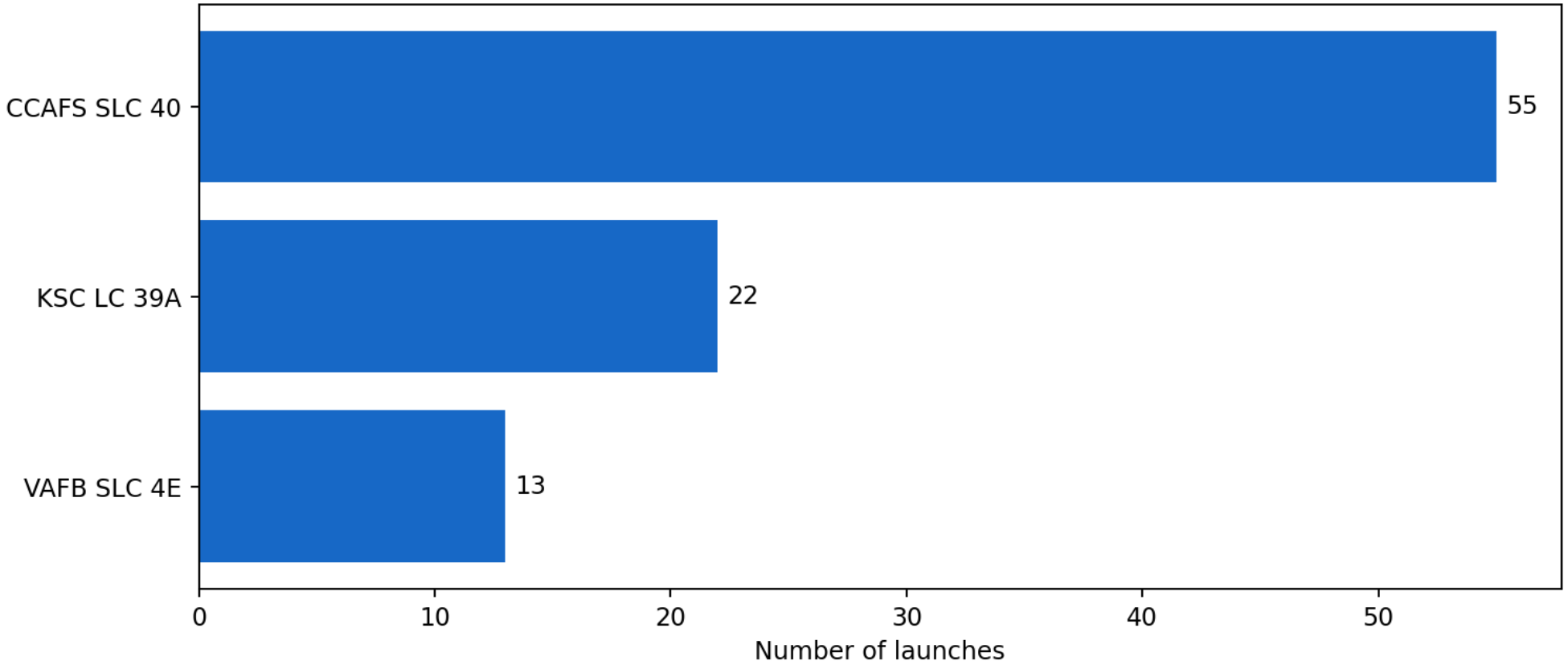
Data wrangling and target construction

- Filtered to 90 Falcon 9 launches with a binary landing label.
- Class = 1: successful first-stage landing; Class = 0: otherwise.
- Imputed missing payload masses; retained launch site, orbit and booster attributes.
- Prepared 80 model features in the completed feature-engineering lab (one-hot encoding).
- Dataset_part_2.csv contains the 90 labelled rows; feature and label rows must remain aligned.

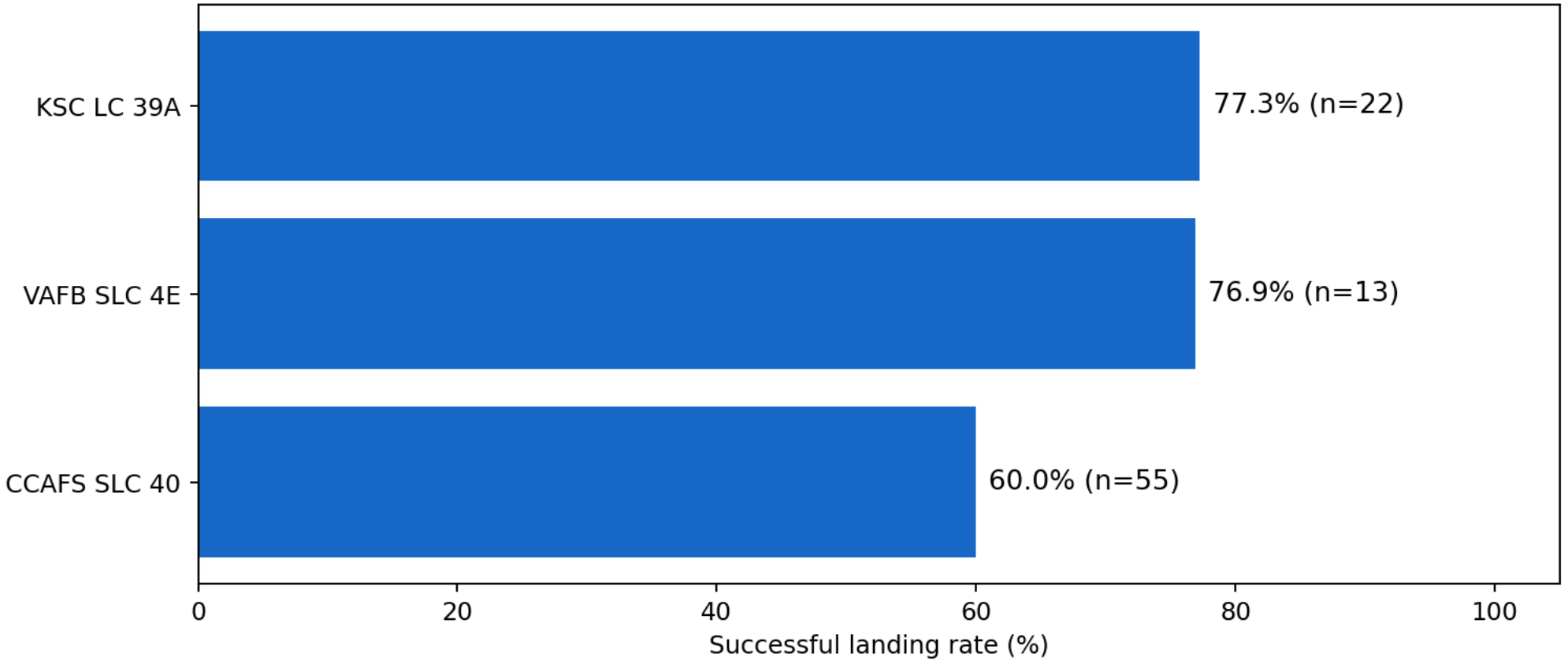
Dataset profile: 90 launches



Exploratory data analysis: launch-site distribution

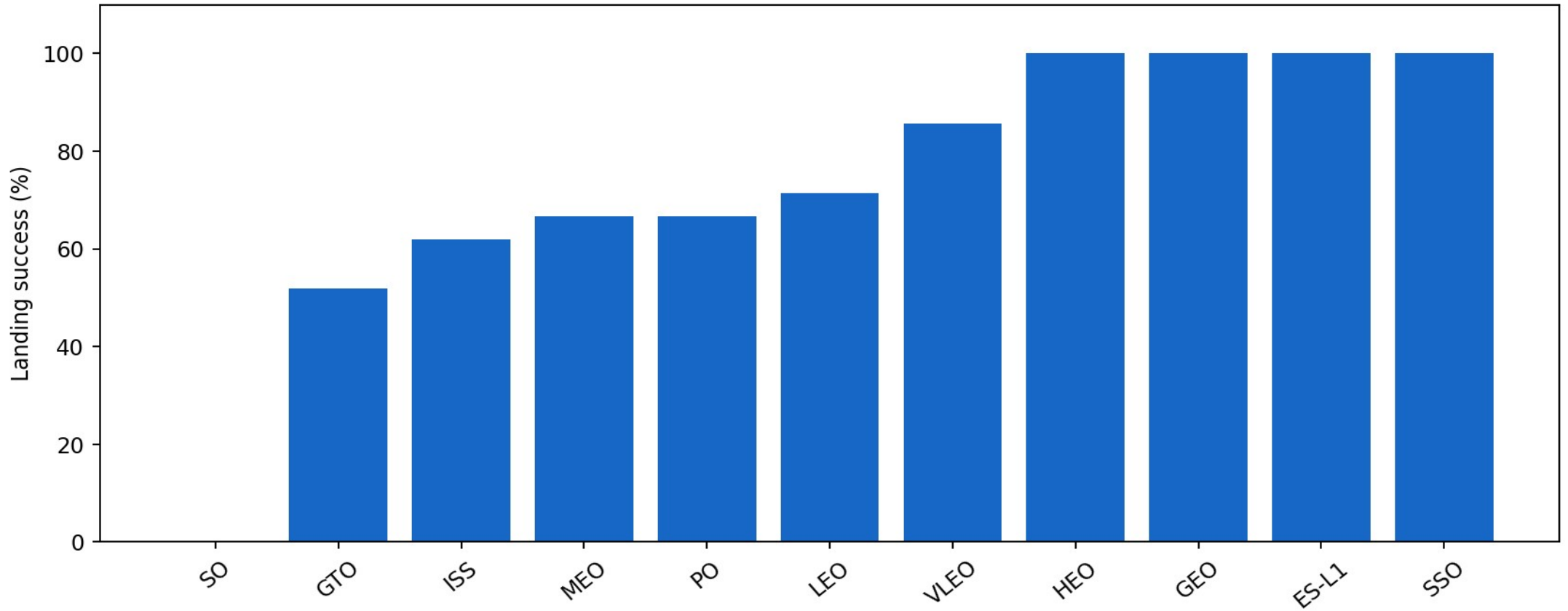


EDA: landing success by launch site

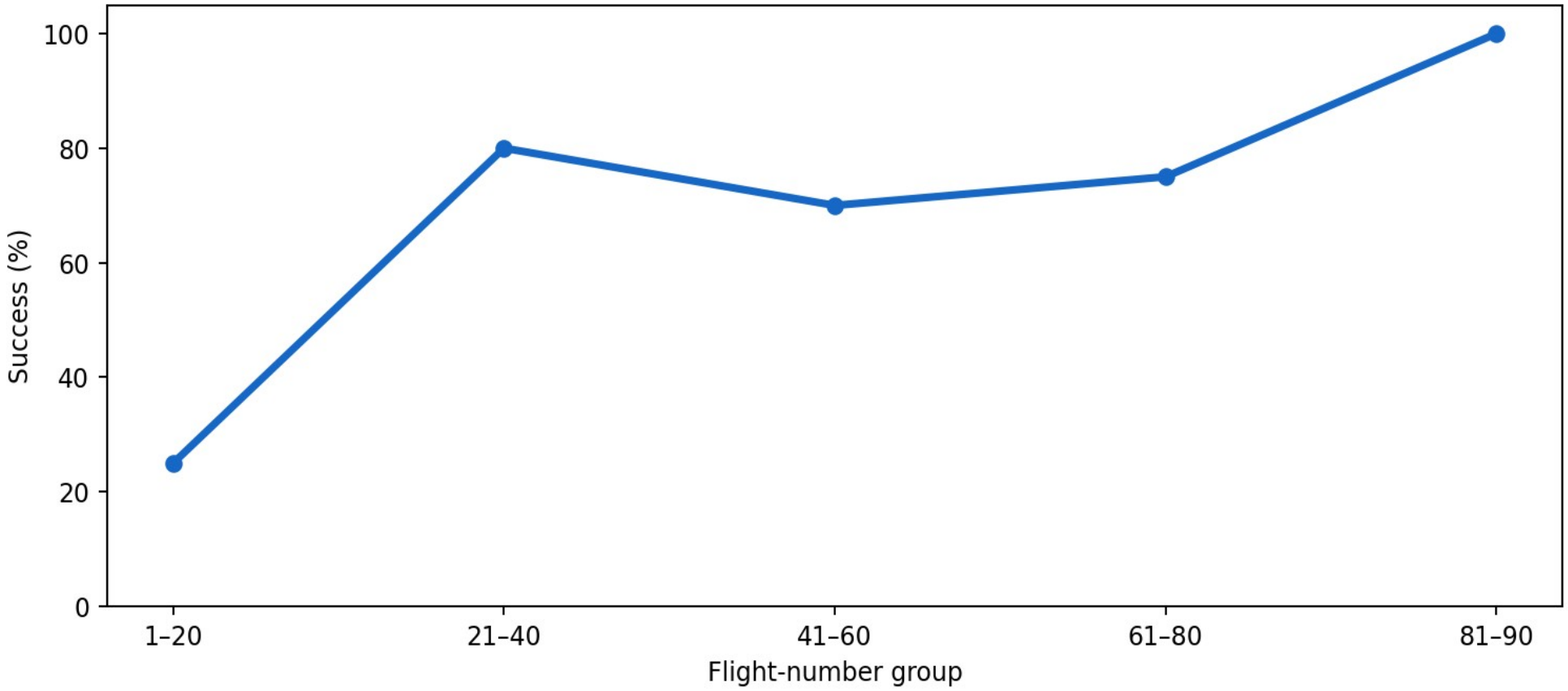


EDA: landing success by orbit

Rates are descriptive; some orbit categories contain very few launches



EDA: landing success across flight history



SQL exploratory analysis

Notebook evidence and key analytical questions

- SQL lab uses a local SQLite database to query Falcon 9 launch records.
- Analytical tasks include launch sites, payload mass, booster versions, landing outcomes and dates.
- The submitted SQL notebook contains query execution, but its saved outputs are not sufficient to verify every requested numeric answer.
- For reproducibility, upload the SQL notebook and database/CSV inputs, then rerun all queries before submission.

Folium: geographic analysis of launch sites

- Mapped launch-site coordinates and annotated their locations.
- Used outcome-coloured markers and MarkerCluster to explore successful and unsuccessful landings.
- Geographic context helps assess coastal proximity and differences between launch sites.
- Distance-to-city/rail/highway sections require verified destination coordinates; do not report unsupported distances.
- The interactive HTML map/notebook should be included in GitHub.

Plotly Dash: an interactive landing dashboard

- Built and ran a working local Dash app at http://127.0.0.1:8050/.
- Launch-site dropdown updates the successful-landing pie chart.
- Payload-range slider filters a scatter plot; booster versions provide additional context.
- Dashboard lab uses a separate 56-row dataset — its percentages must not be mixed with the 90-row modelling dataset.
- Commit spacex-dash-app.py, its source CSV and a screenshot of the running dashboard.

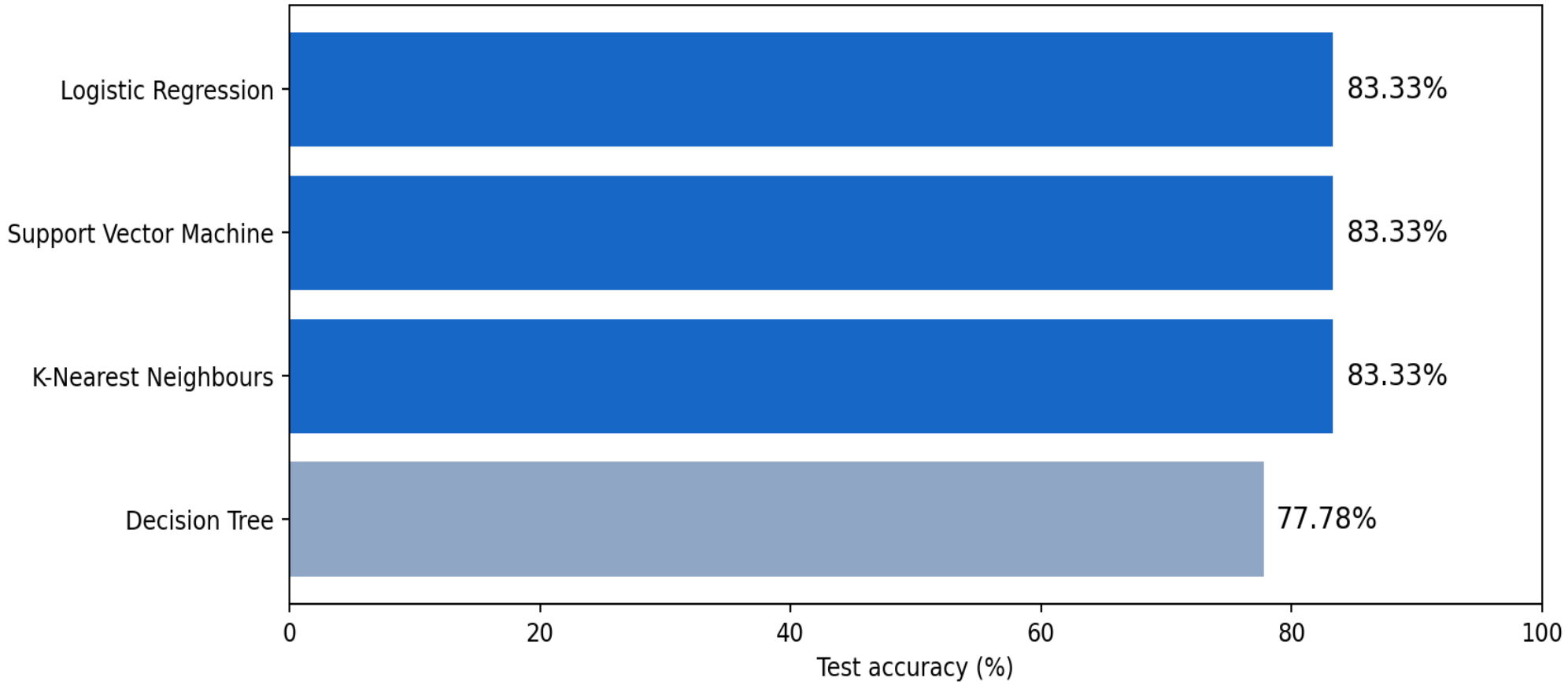
Dashboard findings: 56-launch sample

- KSC LC-39A: 10 successful landings in 13 launches (76.92%).
- Best observed payload band: 2,000–4,000 kg (61.90% landing success).
- 6,000–8,000 kg: 0% observed success in this small sample.
- Booster B5: 100% observed success, but based on only one flight — not a robust estimate.
- These findings describe the dashboard dataset, not the full 90-flight modelling dataset.

Predictive methodology: reproducible classification

- Target Y: Class; predictors X: encoded and standardized launch attributes.
- Train/test split: 80% training (72 flights), 20% test (18 flights), random_state = 2.
- GridSearchCV with 10-fold cross-validation selects model hyperparameters.
- Models: Logistic Regression, Support Vector Machine, Decision Tree and KNN.
- Test accuracy and confusion matrices assess out-of-sample performance.

Model comparison: held-out test accuracy



Confusion matrices: what the predictions missed

- Logistic Regression, SVM and KNN: 12 true positives, 3 true negatives, 3 false positives, 0 false negatives.
- Each correctly classified 15 of 18 held-out flights (83.33%).
- Decision Tree: 11 true positives, 3 true negatives, 3 false positives, 1 false negative.
- Decision Tree correctly classified 14 of 18 flights (77.78%).
- Main error pattern: predicting a landing when the booster did not land (false positive).

Interpretation and limitations

- Flight history, launch site, orbit and payload show useful associations with landing outcomes.
- A three-way model tie means no single best model can be established from accuracy alone.
- Only 18 test launches: one additional mistake changes accuracy by 5.56 percentage points.
- Model features were standardized before the split in the lab; a production-quality rerun should fit scaling on training data only.
- Historical observational patterns do not prove causality or future operational reliability.

Conclusions and recommendations

- Built a complete analytical workflow from collection and cleaning to interactive dashboards and ML.
- Landing success improved over the historical flight sequence in the 90-launch dataset.
- Three classifiers tied at 83.33% on the 18-launch test set; inspect false positives before deployment.
- Next steps: collect more recent flights, avoid data leakage, compare precision/recall and calibrate predicted probabilities.
- Use the dashboard for exploration, not as a validated real-time prediction service.

GitHub portfolio and reproducibility

- Project repository: https://github.com/eokorouka-jpg/SpaceX-Falcon9-Data-Science-Capstone
- Public repository created: SpaceX-Falcon9-Data-Science-Capstone.
- Repository contains the API, Wikipedia, SQL, EDA, Folium and machine-learning notebooks.
- CSV inputs, README.md and the PDF report have been uploaded; verify dashboard scripts and screenshots.
- Keep source data, figures and notebook outputs organised so a reviewer can reproduce the work.

Grading checklist: 15-point submission

- Included: executive summary, introduction and data collection/wrangling methods.
- Included: EDA visualisations, SQL methodology, Folium and Dash analytics.
- Included: predictive modelling methodology, tuned-model results and conclusion.
- Included: original figures, limitations, comparative insights and direct GitHub repository link.
- Before submission: verify SQL/map evidence and ensure the uploaded PDF matches this revised version.
- Submit the accompanying PDF as the assessed presentation.

Appendix: source files and dataset definitions

- dataset_part_2.csv — 90 rows × 18 columns; labelled modelling data.
- Model feature matrix — 90 rows × 80 encoded predictors (as shown in the completed lab).
- spacex-dash-app.py — interactive Dash dashboard; 56-row dashboard sample.
- jupyter-labs-eda-sql-coursera_sqllite.ipynb — SQL lab.
- jupyter-labs-webscraping.ipynb — Wikipedia scraping lab.
- lab_jupyter_launch_site_location.ipynb — Folium geographic lab.